

Yield Excursion — Root-Cause Triage Report

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tool configuration	full
iterations	14 (end_turn)
queries issued	38 (30 filtered, 6 distinct functions)
dimensions reported	SITE_NUM, TEMP_C, VDD

This report is produced from the converted test datalog only. No process, equipment, or maintenance records were available to the analysis. Findings localize a failure within the data; they do not establish a physical cause. Disposition remains an engineering judgement.

Findings

- Overall yield is 95.79% with a measurement failure rate of 0.144% across 3,750 parts (get_overall_yield).
- At TEMP_C=25°C, the overall failure rate is 0.171%, but Sites 3 and 7 show dramatically elevated rates of 0.295% and 0.289% respectively, compared to 0.105% at Site 1 and 0.089% at Site 8 (get_yield_summary grouped by SITE_NUM with TEMP_C=25 filter).
- Site 3 at TEMP_C=25°C and VDD=0.85V has a failure rate of 0.632%, with 10 failures out of 467 parts tested (get_overall_yield with filters TEMP_C=25, VDD=0.85, SITE_NUM=3).
- Site 7 at TEMP_C=25°C and VDD=0.85V has a failure rate of 0.418%, with 6 failures out of 445 parts tested (get_overall_yield with filters TEMP_C=25, VDD=0.85, SITE_NUM=7).
- By contrast, Site 1 at the same conditions (TEMP_C=25°C, VDD=0.85V) has a failure rate of only 0.189% with 3 failures out of 468 parts (get_overall_yield with filters TEMP_C=25, VDD=0.85, SITE_NUM=1).
- Site 8 at TEMP_C=25°C and VDD=0.85V has the lowest failure rate of 0.067% with only 1 failure out of 440 parts (get_overall_yield with filters TEMP_C=25, VDD=0.85, SITE_NUM=8).
- At TEMP_C=25°C and VDD=0.75V, Sites 3 and 7 show only 0.128% and 0.264% failure rates respectively, well below their 0.85V performance (get_fail_rate_by_condition with filters TEMP_C=25, SITE_NUM=3 and SITE_NUM=7).
- vmín_core is the primary failing test at TEMP_C=25°C with a 0.293% failure rate, followed by sram_bist at 0.24% and core_fmax at 0.213% (get_failures_by_test with filter TEMP_C=25).

Ruled Out

- Wafer-level effects: While W01 shows elevated overall failure (0.191%), the worst performance at TEMP_C=25°C, VDD=0.85V is site-dependent. W01 at Site 3, 25°C, 0.85V is not uniquely elevated compared to other wafers at the same site and conditions (get_overall_yield shows site effects are primary).
- Lot-level effects: L45 shows 0.164% measurement failure rate, only marginally higher than other lots. Failures are not concentrated in a single lot (get_yield_summary grouped by LOT_ID).
- Spatial patterns alone: While edge and certain octants show elevated rates overall (0.168% for edge, 0.2% for W octant), the dramatic failures are specifically at Sites 3 and 7 during 25°C, 0.85V testing, not tied to a consistent wafer position pattern (get_yield_by_region shows spatial variation is secondary to site and condition effects).

Assessment

The yield excursion is caused by a three-way interaction between test site (Sites 3 and 7), test temperature (25°C), and supply voltage (0.85V). This is not a wafer, lot, or fundamental spatial manufacturing defect. The data shows Sites 3 and 7 have systematically higher failure rates at 25°C across all voltages, but the effect is most severe at VDD=0.85V. Site 3 at 25°C, 0.85V reaches 0.632% failure rate (10 failures per 467 parts), while comparable sites (1 and 8) achieve 0.189% and 0.067% respectively. This pattern is consistent with a test system calibration or thermal chamber issue specific to Sites 3 and 7 that manifests under high stress conditions (high voltage at room temperature). The magnitude of the effect is substantial and measurable across hundreds of test instances.

Recommended Next Step

Investigate the temperature control and voltage delivery systems at test sites 3 and 7 during 25°C operations. Verify the thermal chamber setpoint accuracy, air flow distribution, and voltage regulation for the 0.85V supply at these two sites. Compare the actual measured voltage and temperature values logged during testing at these sites versus other sites. This is likely an ongoing condition requiring equipment maintenance or recalibration before these sites return to normal yield performance.

Appendix — Queries Issued

Every query made during the investigation, in order. Included so that each finding above can be traced to the data that produced it.

#	iter	query	restricted to
1	1	get_overall_yield	—
2	1	get_failures_by_test	—
3	2	get_yield_summary (group_by=LOT_ID)	—
4	2	get_yield_summary (group_by=WAFER_ID)	—
5	2	get_yield_summary (group_by=SITE_NUM)	—
6	3	get_fail_rate_by_condition	—
7	3	get_yield_by_region	—
8	4	get_yield_summary (group_by=WAFER_ID)	SITE_NUM=7
9	4	get_yield_summary (group_by=WAFER_ID)	SITE_NUM=3
10	4	get_yield_summary (group_by=LOT_ID)	WAFER_ID=W01
11	5	get_fail_rate_by_condition	WAFER_ID=W01
12	5	get_fail_rate_by_condition	SITE_NUM=7
13	5	get_fail_rate_by_condition	SITE_NUM=3
14	6	get_overall_yield	TEMP_C=25
15	6	get_overall_yield	TEMP_C=25, WAFER_ID=W01
16	6	get_overall_yield	TEMP_C=25, SITE_NUM=7
17	6	get_overall_yield	TEMP_C=25, SITE_NUM=3
18	7	get_overall_yield	TEMP_C=25, WAFER_ID=W01, SITE_NUM=7
19	7	get_overall_yield	TEMP_C=25, WAFER_ID=W01, SITE_NUM=3
20	7	get_yield_by_region	TEMP_C=25
21	8	get_failures_by_test	TEMP_C=25
22	8	get_failures_by_test	WAFER_ID=W01
23	8	get_failures_by_test	SITE_NUM=7
24	9	get_distribution_stats (test_txt=vmin_core)	—

#	iter	query	restricted to
25	9	get_distribution_stats (test_txt=vmin_core)	TEMP_C=25
26	9	get_distribution_stats (test_txt=vmin_core)	WAFER_ID=W01
27	10	get_yield_by_region	TEMP_C=25, WAFER_ID=W01
28	10	get_yield_by_region	TEMP_C=25, SITE_NUM=7
29	10	get_yield_by_region	TEMP_C=25, SITE_NUM=3
30	11	get_yield_summary (group_by=SITE_NUM)	TEMP_C=25
31	11	get_fail_rate_by_condition	TEMP_C=25, SITE_NUM=1
32	11	get_fail_rate_by_condition	TEMP_C=25, SITE_NUM=8
33	12	get_fail_rate_by_condition	TEMP_C=25, SITE_NUM=3
34	12	get_fail_rate_by_condition	TEMP_C=25, SITE_NUM=7
35	13	get_overall_yield	TEMP_C=25, VDD=0.85, SITE_NUM=3
36	13	get_overall_yield	TEMP_C=25, VDD=0.85, SITE_NUM=7
37	13	get_overall_yield	TEMP_C=25, VDD=0.85, SITE_NUM=1
38	13	get_overall_yield	TEMP_C=25, VDD=0.85, SITE_NUM=8